

1. Explain Programming and python in detail
• definition and Purpose of Programming

Definition:- Programming is the process of developing and writing a set of instructions using a programming language so that a computer can perform specific tasks

- * To solve problem using computers
- * To perform tasks automatically
- * To develop application, software, websites and system

characteristics and applications of python

characteristics:- python has several important characteristics that make it popular and easy to use

- * Simple and Easy to learn:- python has clear and English-like syntax so beginners can understand it easily.
- * High-level language:- Programmers do not need to manage memory or hardware details.

python handles them automatically.

* interpreted language :- python executes codes line-by-line which makes debugging easier

* object-oriented :- python supports classes and objects, and concepts like inheritance and polymorphism

* platform independent :- python program can run on windows, Linux, and mac without modification

Applications :- Following are the Applications of python

* websites :- used to make websites and web application

* Data Analysis :- used to make study and understand large amount of data.

* machine learning and AI :- used to make smart system like face recognition etc.

* Automation :- used to do repeated tasks automatically

* Games :- used to make simple games

* Cyber security :- used to make software for computers.

Types of comments in python with Syntax

python supports three types of comments

* Single-line comment: used to write a short comment in one line and starts with the # symbol

Syntax :- # This is a single-line comment

* Multi-line comment: used to write a ~~short~~ long comment in multiple lines and written using triple quotes " " " (or) ' ' ' " " "

Syntax :- '''
 this is a
 multi-line comment
 in python
 '''

Importance of python in modern software development
python is very important in today's software world because of its simplicity, powers, and wide usage.

Some key points are:-

- * Easy to learn: python is simple so developers can write program faster
- * Saves time: less code is needed so software is built quickly.

Describe data types and operators in python with suitable example.

Built-in data types in python (Numeric, Sequence, set mapping, Boolean)

Data types:- A data type tells what kind of value is stored in a variable - such as numbers text true/false. etc.

python has different data types

Numeric data types:- Numeric data types are used to store numbers. There are three of numeric datatypes

- int - whole number
- float - Decimal number
- complex number with imaginary part ($a+bi$)

Example:- $x=10$ # int

$y=3.14$ # float

$z=2+3j$ # complex

Sequence data types:- These store multiple items in an ordered manner.

• String (str): stores text. String data types are written in quotes

name = "python"

3. Explain python input and output operations in detail.

* `input()` Function :- `input()` Function are used to take input from the user

Syntax: `var = input ("message")`

It always returns data as string by default.

* Default data type :- Data taken using `input()` is always stored as str (string type)

Type conversion (casting)

To convert input to other data types:

→ convert to integer :-

```
num1 = int (input ("Enter a number"))
```

→ convert to float:

```
Price = float (input ("Enter price :"))
```

→ convert to boolean (manually interpreted)

Example : `x = input ("Enter anything: ")`

```
print (type(x)) # output : <class 'str'>
```

Key points:-

- * `input()` function pauses program & wait for user input
- * Useful for interactive program
- * Requires type conversion for mathematical operations.

11. Discuss control statements and Decision-making statements in Python

• meaning and importance of control statements

meaning :- control statements are special instruction in a programming language that control the flow of execution of a program they decide which part of the code runs, how many times it runs and under what conditions it runs.

Types of control statement

In Python control statement are mainly of three types they are.

conditional (Decision-making statements):-

used to check conditions and decide what to do

- if
- if-else
- elif

→ Looping (Repetition) Statement:-

used to repeat a block of code multiple times

- for
- while.

→ Jump Statement:-

used to change the normal flow inside loops

- break - , • continue , return.

write an essay on python programming fundamentals.

- Role of Programming in problem solving
Programming plays an important role in solving problems because it helps us tell the computer what to do in a clear and step-by-step ways

- Python Syntax simplicity and readability.
Python is known for having a simple and clear syntax. This means the way we write python code is

- easy to understand
- easy to write
- close to normal english language

1. Program on movie ticket pricing

```
age = int(input("Enter your age:"))
```

```
is3D = int(input("Enter 1 if you are watching a  
3D movie else enter 0:"))
```

```
if is3D == 1:
```

```
    if age < 13:
```

```
        print("Your ticket price is ₹ 200")
```

```
elif age >= 13 and age <= 59:
```

```
    print ("Your ticket price is ₹ 300")
```

```
else
```

```
    print ("Your ticket price is ₹ 250")
```

```
elif age < 13:
```

```
    print ("Your ticket price is ₹ 150")
```

```
elif age >= 13 and age <= 59:
```

```
    print ("Your ticket price is ₹ 250")
```

```
else:
```

```
    print ("Your ticket price is ₹ 200")
```

output:-

Enter your age : 11

Enter 1 if you watching a 3D movie else

enter 0 : 1

Your ticket price is ₹ 200

Program on college Attendance Rule

```
attper = int(input("Enter your attendance percentages"))
```

```
medcor = int(input("Enter 1 if you have medical  
certificate else 0"))
```

```
if attper >= 75 or (attper >= 60 and medcor == 1):
```



```
print ("you allowed to write exam")
```

```
else:
```

```
print ("you not allowed to write exam")
```

output:-

Enter your attendance percentage : 20

Enter 1 if you have medical certificate else 0: 1
you not allowed to write exam.

Program on E-commerce Discount.

```
bilamo = int (input ("Enter your bill amount :"))
```

```
isPrime = int (input ("Enter 1 if you are prime  
member else 0:"))
```

```
if isPrime == 1:
```

```
    if bilamo >= 5000:
```

```
        dis = 0.25 * bilamo
```

```
    elif 2000 <= bilamo <= 4999:
```

```
        dis = 0.15 * bilamo
```

```
    else:
```

```
        dis = 0
```

```
else:
```

```
    if bilamo >= 5000:
```

```
        dis = 0.20 * bilamo
```

```
    elif 2000 <= bilamo <= 4999:
```

```
        dis = 0.10 * bilamo
```

```
    else:
```

dis = 0

amo = bilamo - dis

print ("final amount to be paid is : amo)

output:-

Enter your bill amount : 4500

Enter 1 if you are prime member else 0:0

final amount to be paid is 4050.0

u. Program on Smartphone Battery warning

batper = int (input ("Enter your battery percentage"))

ischar = int (input ("Enter 1 if phone is charging
else 0"))

if ischar == 0 :

if batper <= 20 :

print ("low battery")

elif 21 <= batper <= 80 :

print ("Normal")

else

print ("Full")

elif ischar == 1 :

print ("charging")

else

print ("Enter 0 or 1")

output:-

Enter your battery percentage : 87

enter 1 if phone is charging else 0: 1
charging

5. program on Driving License check

```
age = int(input("Enter your age :"))
```

```
testpassed = int(input("Enter 1 if you passed  
the test else 0:"))
```

```
if (age >= 18 and testpassed == 1) or age >= 60:  
    print("Eligible")
```

```
else
```

```
    print("Not Eligible")
```

output:-

Enter your Age: 60

Enter 1 if you passed the test else 0: 1
Eligible.